Kyle Tran

Mechanical Engineering, Robotics minor | Northeastern '29 | Seeking a Summer 2027 hardware or robotics co-op
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EDUCATION

Northeastern University | *Boston, MA* Expected May 2029
Bachelor of Science in Mechanical Engineering, Minor in Robotics | GPA: 3.87/4.00
Relevant Coursework: Cornerstone of Engineering I, Statics, Thermodynamics, Materials Science, Probability and Statistics

PROJECTS

D.R.E.A.M.S Autonomous Environmental Survey Boat | *Co-founder, with Matthew Hong* Jan 2026 – Present
- Design lead on v3 of a twin-hull survey boat; on v2, built and fiberglassed the hull and did all of the electronics
- Built v1 in one semester in London (I did the bridge, propulsion and winch): it navigated to GPS waypoints on its own at St Katharine Docks over a 1 km radio link; we showed it at NU London's first Creators' Showcase | ka1e27.github.io/dreams
- Designed the v3 KiCad boards, a 4-layer, 140-part carrier for a Raspberry Pi 5 and two Pico 2s with CAN motor control plus a sensor-pod board, and got PCBWay to sponsor their fabrication
- Ran CFD on the v3 hull in SolidWorks Flow Simulation: about 13 N of drag at 2.5 knots, within 20% of our sizing model
- Redesigned the system for v3 after v2's bench tests exposed motor-control, power and safety flaws; specified the architecture and directed AI coding agents to implement the software

Combat Robotics | *Team Dihydrogen Monoxide* Mar 2023 – Jun 2025
- Designed and built about ten robots across five weight classes in Onshape: spinner and drum weapons, brushless drive, aluminum frames sent out for CNC machining, 3D-printed nylon and polycarbonate armor
- 4th nationally in Plastic Antweight (2024 RCL National Championship); U.S. rankings of top-15 (1 lb), top-30 (1 lb plastic) and top-35 (3 lb); 4 first- and 8 second-place regional finishes
- Designed a single-cam shuffler drive so the newest version of Achilles (1 lb) walks instead of rolling

Autonomous Maze-Navigating Robot | *Cornerstone of Engineering I, with Matthew Hong* Oct 2025 – Dec 2025
- Designed a 3D-printed tank-drive chassis in Onshape to replace the class kit, which tipped in fast turns: 1.6 in tall, about half the kit's height, and every printed part fit the first time
- Wrote the MicroPython firmware: proportional control that centers the robot between walls using two angled ultrasonic sensors, an LCD program menu, and asyncio so animations never stall the motors
- Added crowned wheels to keep the rubber-band treads on in hard turns; completed all three challenge courses

Adaptive Compliant Robotic Arm | *Personal Project, In Design* 2026 – Present
- Five-axis arm; three cycloidal gearboxes (14:1 to 22:1) with output-side encoders, designed to hold ±0.25°
- Built a gearbox simulator with AI coding agents; all three gearboxes pass its stress and bearing-life checks

LEADERSHIP EXPERIENCE

President | *Combat Robots Club, Burlingame, CA* Oct 2023 – Jun 2025
- Taught CAD, soldering, electronics and 3D printing; secured sponsorships and ran club finances and event travel
- Led the club to 16 podium finishes (4 first, 7 second, 5 third) across 5 events

Subsystem Lead | *Iron Panthers, FRC Team 5026, Burlingame, CA* Sep 2021 – May 2024
- Designed and prototyped the spindexer mechanism in CAD; led 10+ students, advised by 5 professional mentors

Ceramic Tools R&D | *Burlingame High School* Oct 2023 – May 2025
- Designed, 3D-printed and published 10+ tools for the ceramics classes, free for anyone to print on MakerWorld

SKILLS

CAD and design: SolidWorks, Onshape, Fusion 360; CFD, mechanism design, DFM, rapid prototyping
Fabrication: FDM 3D printing, composite layup (fiberglass, carbon fiber), laser cutter, PCB mill, soldering; beginner: 2.5D CNC machining, CNC router, waterjet
Electronics: embedded systems, RP2040 microcontrollers, Raspberry Pi, KiCad PCB design, motor control, sensors
Programming: Python, MicroPython (asyncio), MATLAB, Java, GitHub; AI coding agents (Claude Code)

AWARDS

4th, Robot Combat League National Championship, Plastic Antweight (2024) | 1st, CROS March Madness (2024) | 1st, BCRC Bell Brawls IV (2024) | 1st, Spartan Bell Slugfest (2024) | 1st, NorCal Robotics Expo (2023)